

Sequential Models

1. **RNN**
2. **LSTM**
3. **GRU**

1. RNN (recurrent NN)

Great for modeling sequential data.

Eg of sequential data - speech, audio, text etc

Use cases - speech recognition, language translation, stock prediction etc

Example of a sequential memory

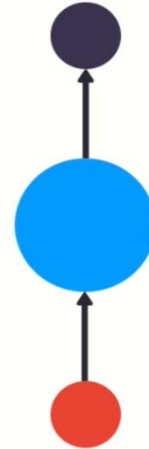
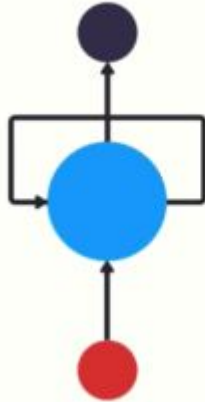
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

ZYXWVUTSRQPONMLKJIHGFEDCBA

H I J K L M N O P Q R S T U V W X Y Z

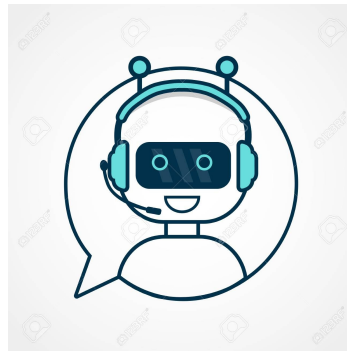
THAT MEANS SEQ. DATA IS EASIER TO REMEMBER!
& RNNs ARE ABSTRACT CONCEPT OF SEQ. MEMORY

RNN



CHATBOT EXAMPLE

YOU: WHAT TIME IS IT?

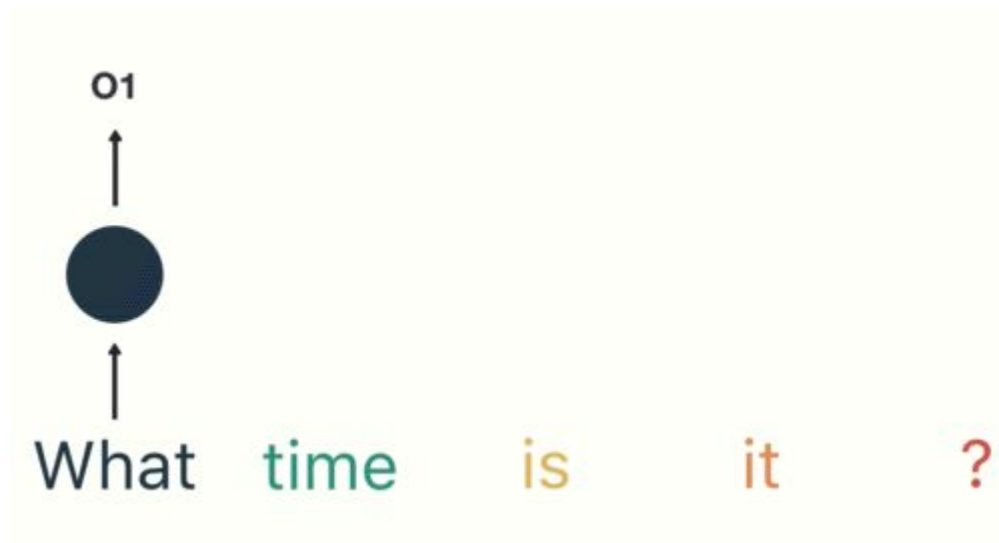


ASKING FOR TIME

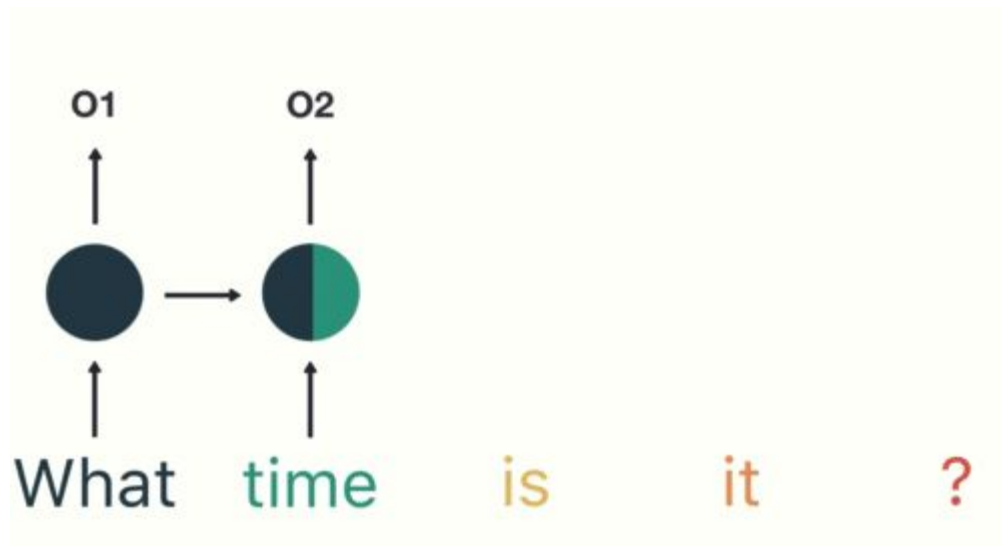
What time is it?

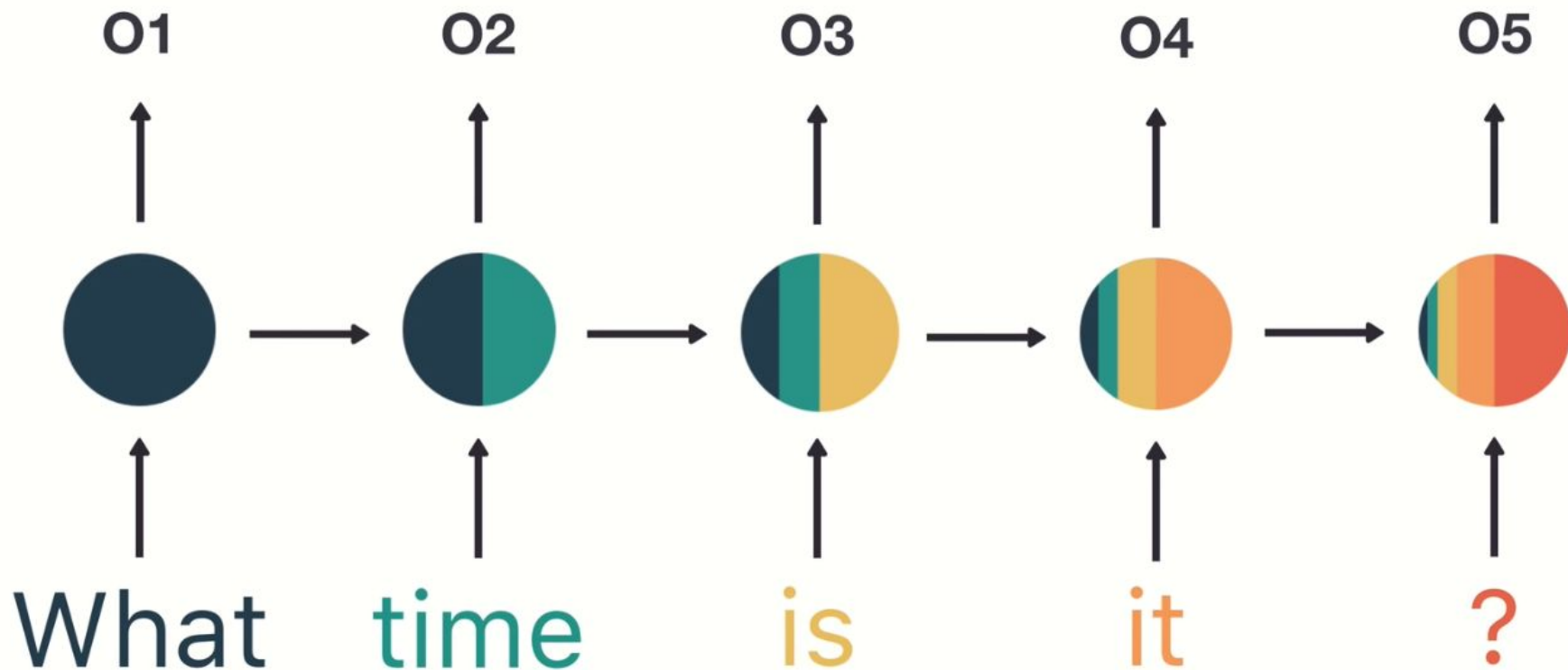
What time is it ?

What time is it ?

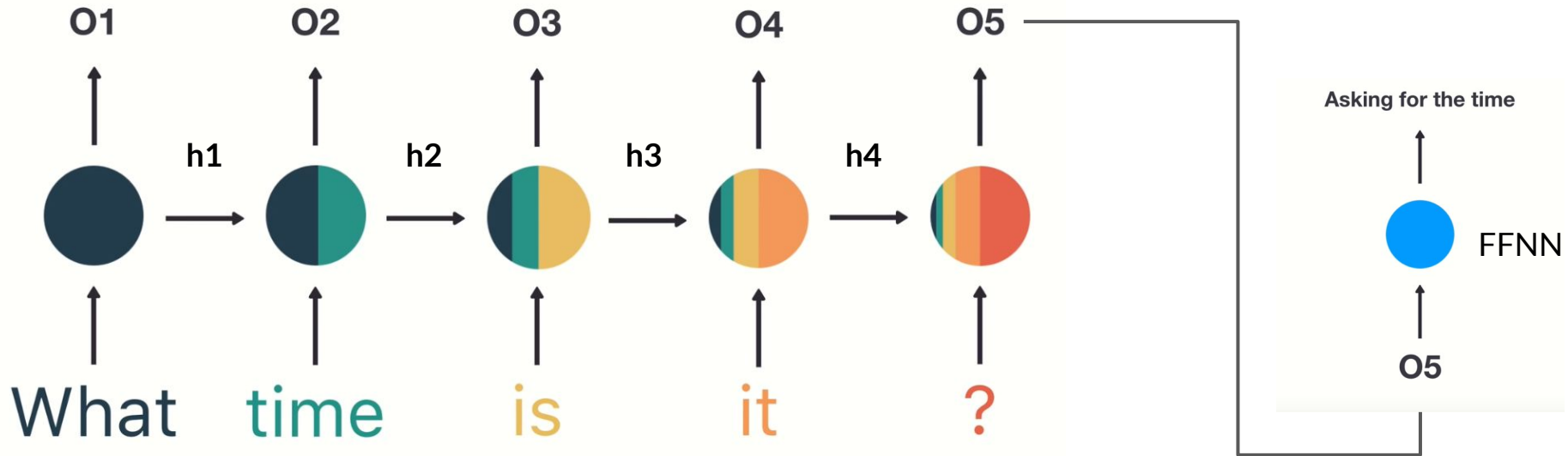


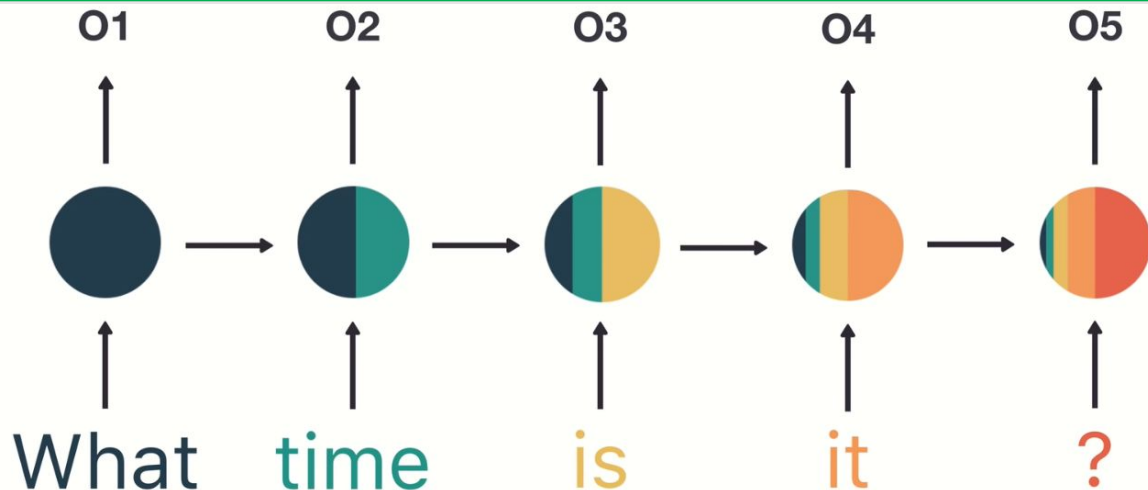
RNN





RNN





RNNs suffer from vanishing gradient hence they have short term memory.

$$\text{new weight} = \text{weight} - \text{learning rate} * \text{gradient}$$

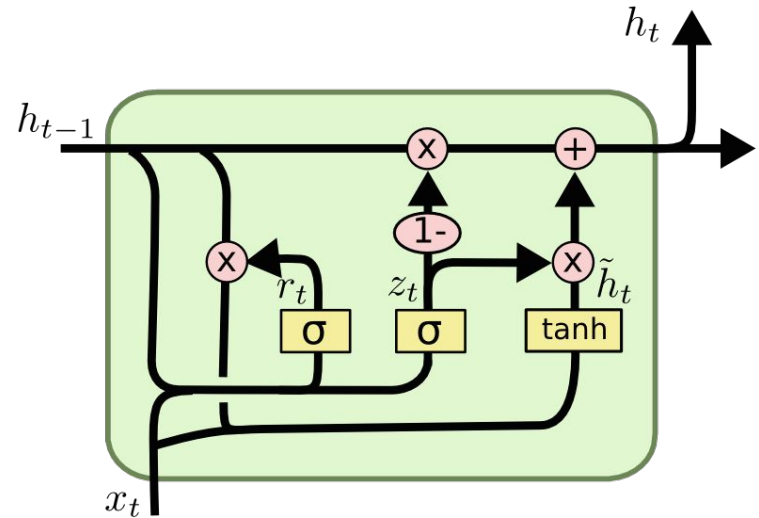
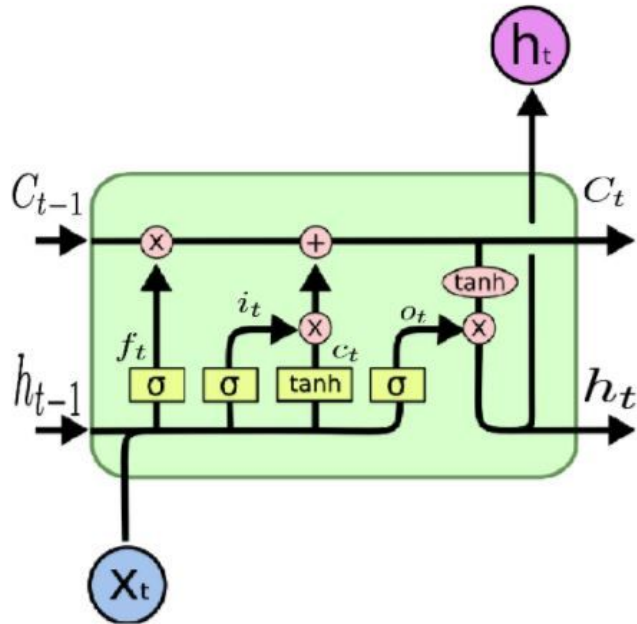
$$\boxed{2.0999} = \boxed{2.1} - \boxed{0.001}$$

Not much of a difference

update value

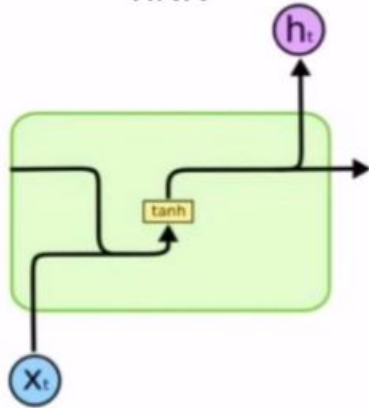
LSTM & GRU

LSTM and GRU (comes to rescue to tackle short term memory issue)



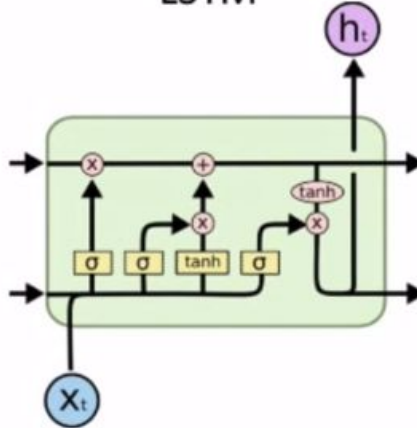
LSTM & GRU

RNN



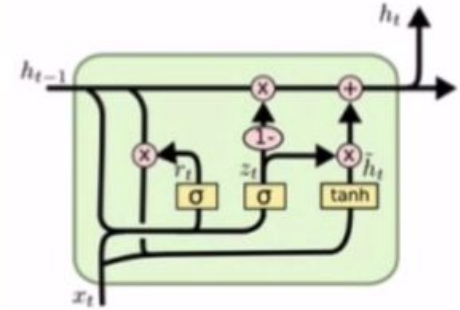
1. Simple arch.
2. Good for speed as it has less tensor ops.
3. Suffers short term memory.

LSTM



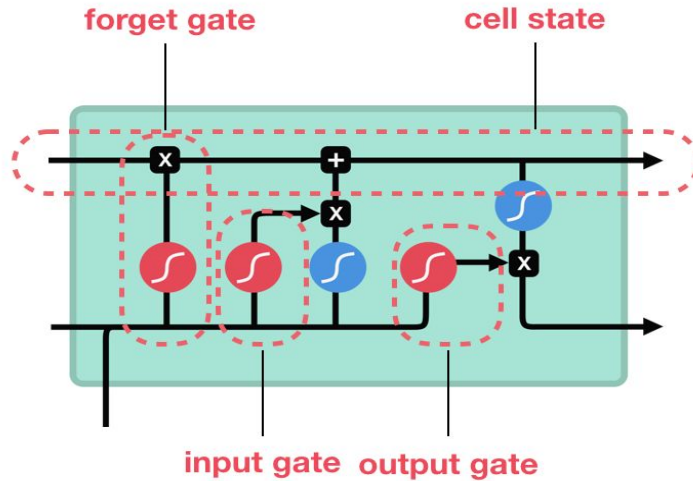
1. Quite complex.
2. Good for modeling longer sequences which has long term dependencies.
3. Conquers short term memory.

GRU

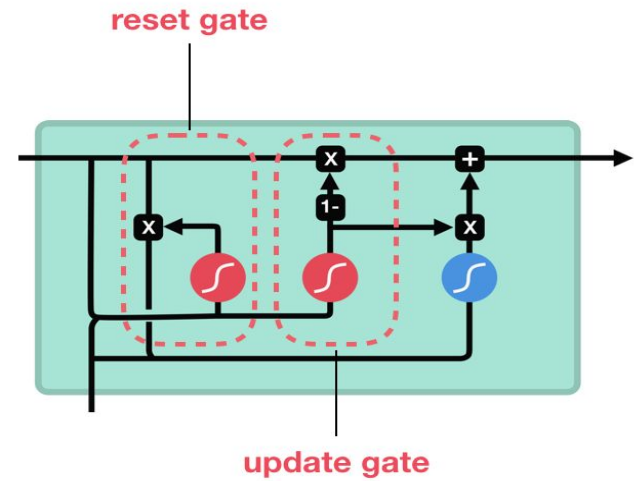


LSTM & GRU

LSTM



GRU



sigmoid



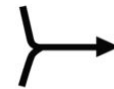
tanh



pointwise
multiplication



pointwise
addition



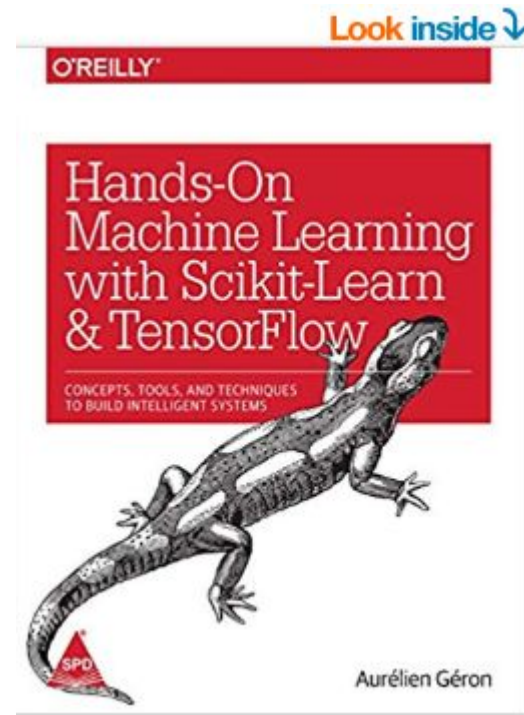
vector
concatenation

Customer review example-

5.0 out of 5 stars

A perfect book for ML Scikit and Tensorflow

This is one of the best books you can get for someone who is just starting out in ML, in its libraries such as Tensorflow, *It covers the basics very good.* As a book, it is 5/5

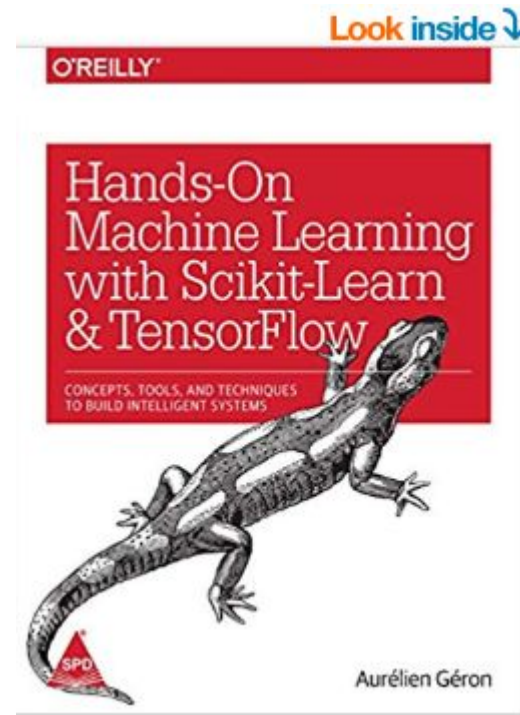


Customer review example-

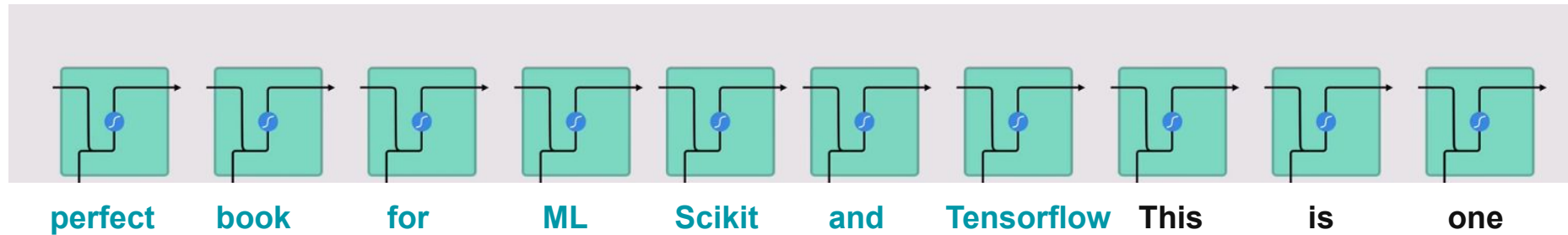
5.0 out of 5 stars

A perfect book for ML Scikit and Tensorflow

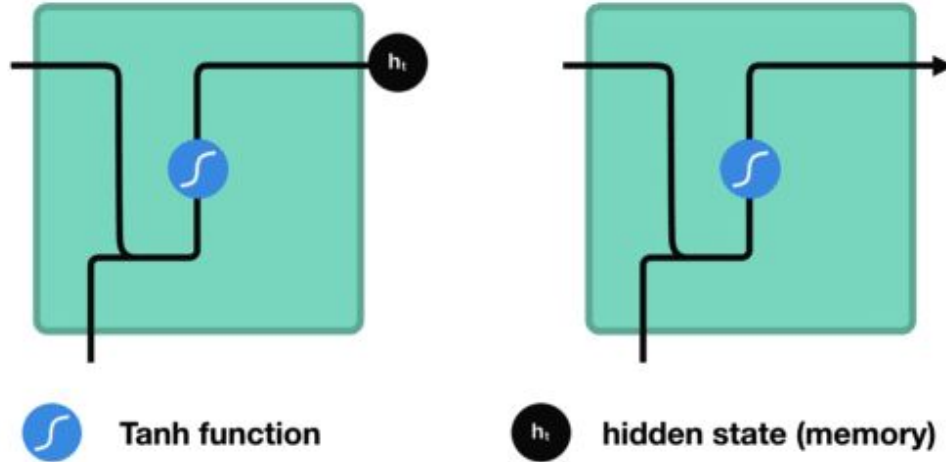
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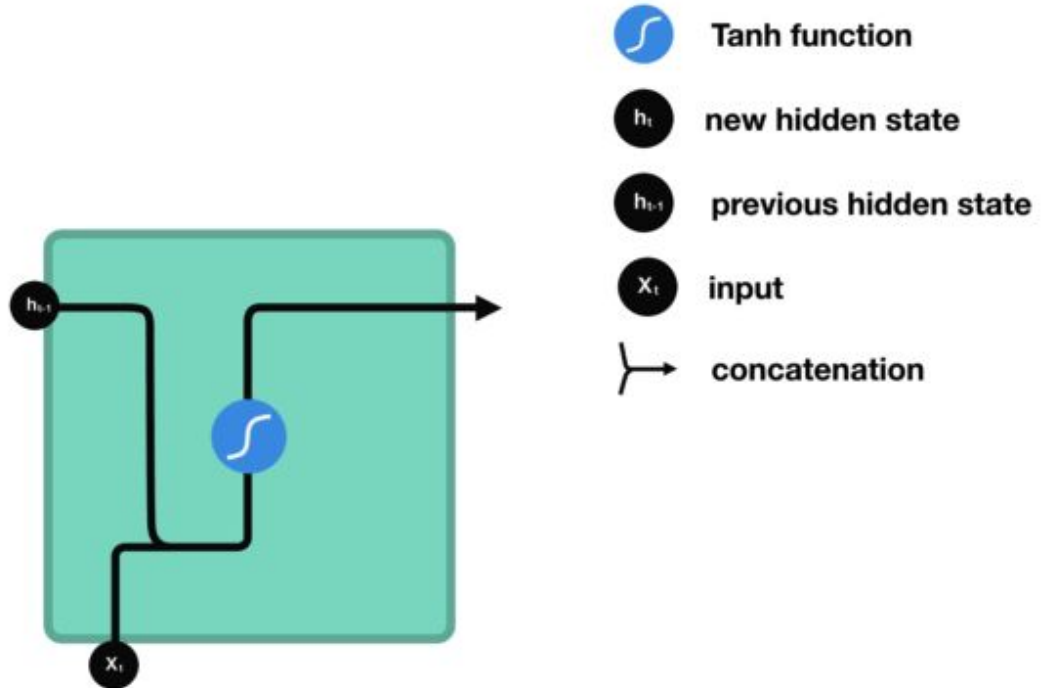


RNN



LSTM

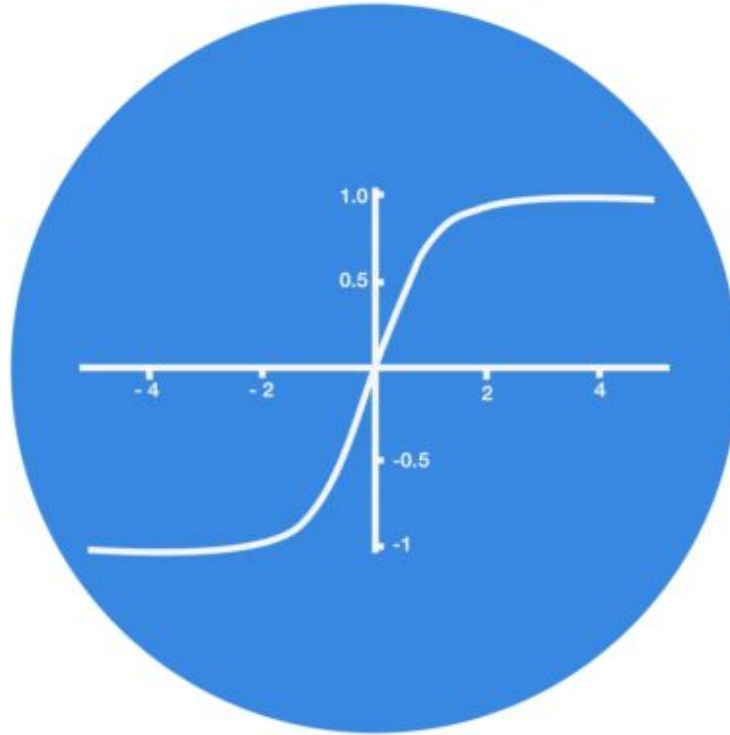




Tanh function and Sigmoid function

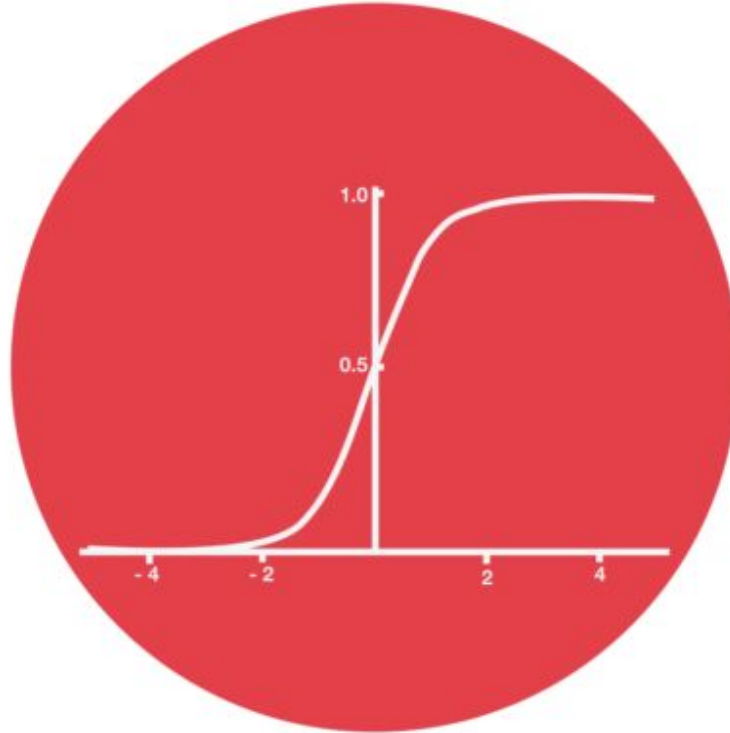
Activation Used : Tanh

5
0.1
-0.5



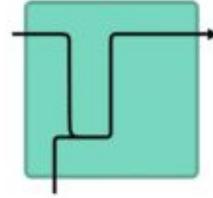
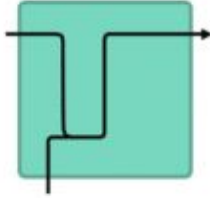
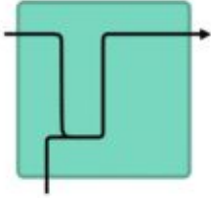
Activation Used : Sigmoid

5
0.1
-0.5

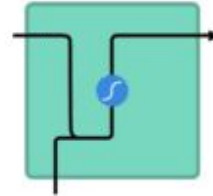
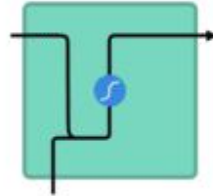
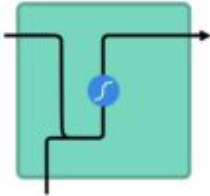
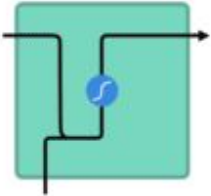


Activation Used

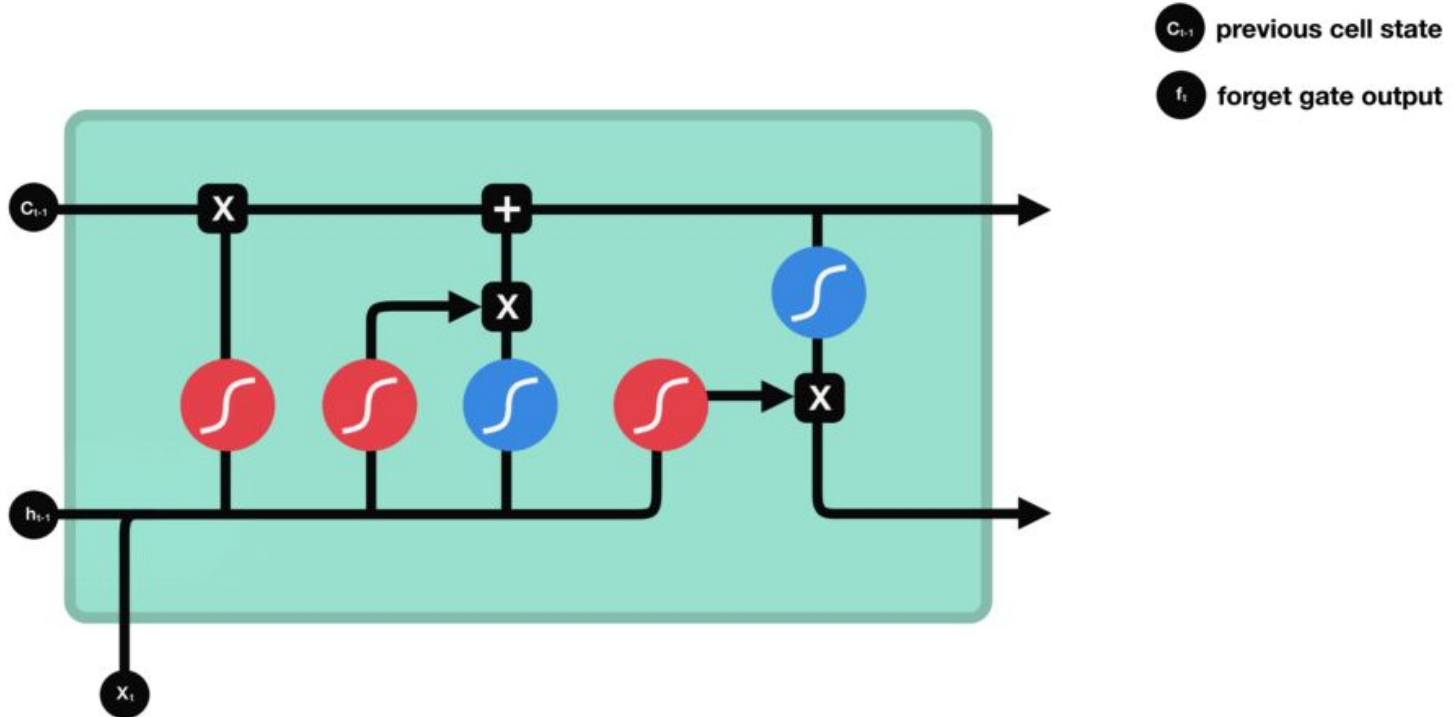
5
0.01
-0.5



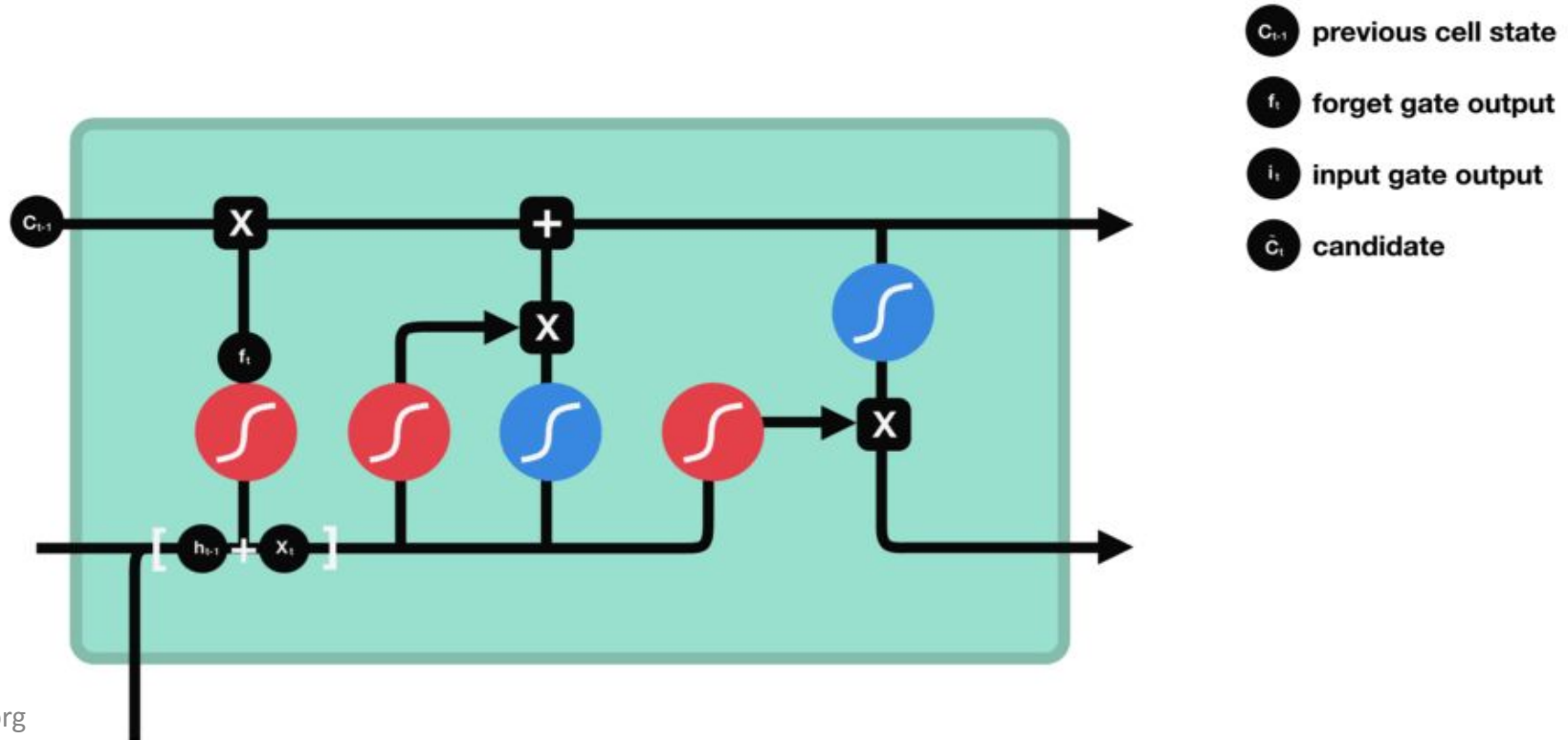
5
0.01
-0.5



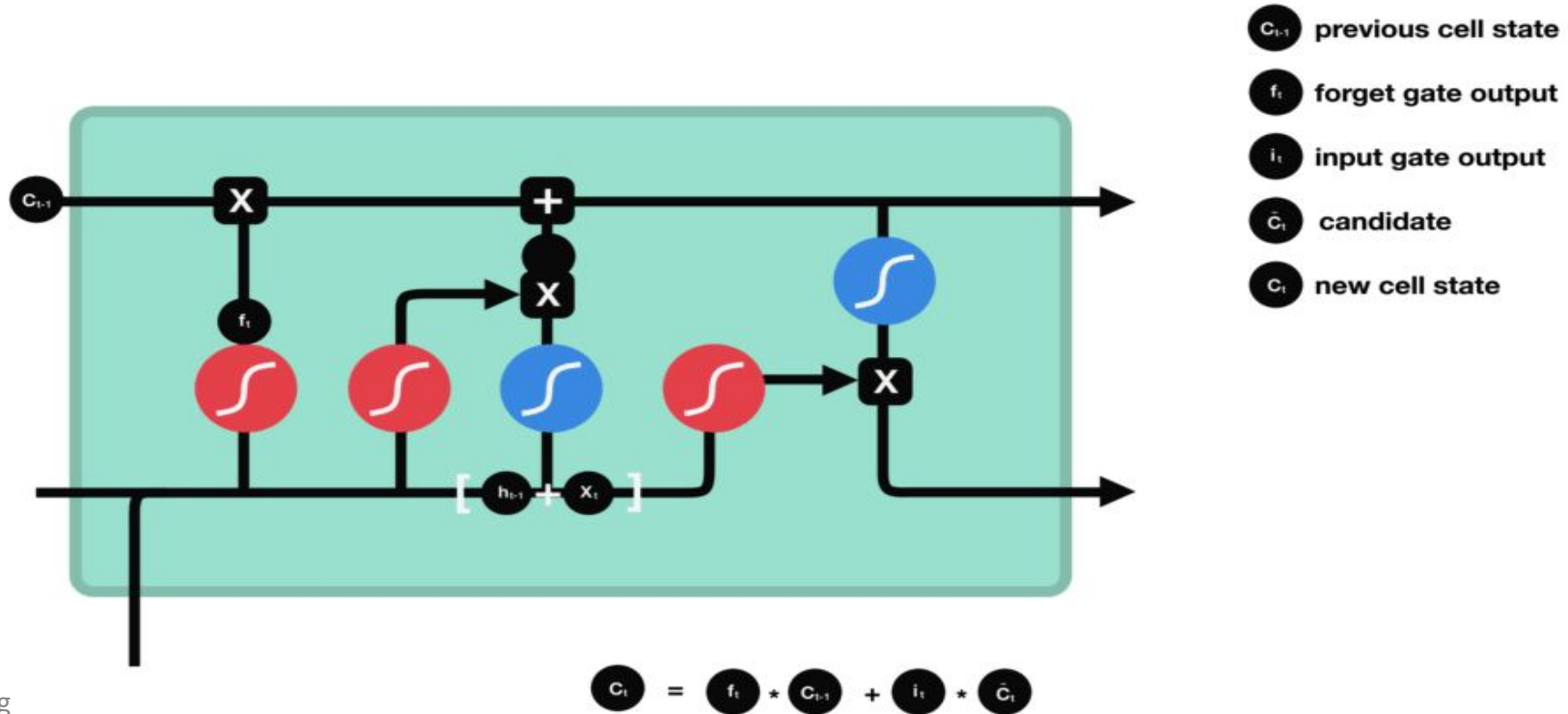
Forget gate



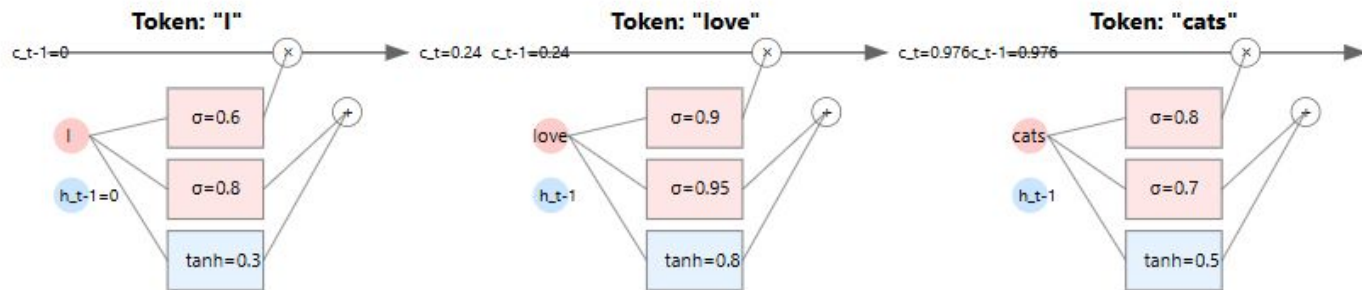
Input Gate



Cell State



LSTM Processing Sequence



Token "I":

Old info: $0 \times 0.6 = 0$
 New info: $0.3 \times 0.8 = 0.24$
 Final: $0 + 0.24 = 0.24$

Token "love":

Old info: $0.24 \times 0.9 = 0.216$
 New info: $0.8 \times 0.95 = 0.76$
 Final: $0.216 + 0.76 = 0.976$

Token "cats":

Old info: $0.976 \times 0.8 = 0.78$
 New info: $0.5 \times 0.7 = 0.35$
 Final: $0.78 + 0.35 = 1.13$

 Sigmoid Gates (0 to 1)

 Tanh (-1 to 1)

 Operation Points ($\times$ or $+$)

Output Gate

